Jeevana Priya Inala
Resume

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Education

2012–Present **Bachelor of Science**, *Massachusetts Institute of Technology*, *GPA – 4.9*.

Relevant Coursework Foundations of Program Analysis; Machine Learning; Computer Systems Security; Database Systems; Multicore Programming; Automata, Computability and Complexity; Performance Engineering;

Experience

Undergraduate Research

Oct 2014–Present **Type-Aware Transactions for Faster Concurrent Code**, CSAIL, MIT.
Supervisors: Prof. Barbara Liskov, Prof. Eddie Kohler

Implemented a software transactional object (STO) system that tracks semantic operations on transactional datatypes rather than low-level memory reads and writes. Evaluated the approach on STAMP benchmarks to illustrate that STO performs better than regular untyped transaction systems like TL2. Under submission.

Dec 2014–Present **Synthesis of optimal constraints for SAT solver**, CSAIL, MIT.
Supervisor: Prof. Armando Solar-Lezama

In this project, we are using SKETCH (a program synthesis tool) to automatically figure out optimal CNF clauses from high level DAGs represented as functions. For this, we formalized this synthesis problem and developed optimal specifications for these kinds of problems. We have used it to synthesize constraints for several boolean functions. We are now working on using this formulation for more complex functions involving bit vectors and integrating these optimal encodings directly into SKETCH.

Sep 2013–July 2015 **Type Assisted Synthesis of Recursive Transformers on Algebraic Data Types**, CSAIL, MIT.
Supervisor: Prof. Armando Solar-Lezama

Implemented SYNTREC (an extension to SKETCH) that can synthesize interesting functions involving algebraic data types and pattern matching. One of the novel aspects of this work is the combination of type inference and counterexample-guided inductive synthesis (CEGIS) in order to support very high-level notations for describing the space of possible implementations that the synthesizer should consider. Under submission. Draft at http://arxiv.org/abs/1507.05527

Internship

Summer 2014 **Software Engineering Intern**, GOOGLE INC., CA.

Designed and implemented the mobile app for Content ID feature. Improved the app with mobile specific characteristics to show that the mobile app is better than the web app.

Awards

2015 First place in PLDI student research competition
2014-2015 Actifio Undergraduate Research and Innovation Scholar
2012 Gold medal at 13th Asian Physics Olympiad, India
2011 Gold medal and Best in Theory in 5th International Olympiad in Astronomy and Astrophysics, Poland
2012 Silver medal and Asian Girl topper in 43rd International Physics Olympiad, Estonia
2012 Secured rank 21 in IIT Joint Entrance Examination

Skills

Java, C++, C, Python, HTML, CSS, JavaScript, PHP, MySQL, Objective-C
Experienced with JQuery, GIT, Eclipse, Android development.

Course Projects

Dec 2014 **Concolic Execution for Django**, *Course: 6.858 Computer Systems Security*, MIT.

Implemented a concolic execution framework for testing Django applications. Uses a Z3 solver to solve for symbolic path conditions to find the failing test case.

Jan 2013 **BatteCode**, MIT.

The 6.370 Battlecode programming competition is a unique challenge that combines battle strategy, software engineering and artificial intelligence. Implemented Bug algorithm for navigation of soldiers and other strategies like division of soldiers into groups, group navigation and opponent strategy based strategy.